

Lab Assessment Rubrics of Digital System Design (CPE-344)

Lab Number		03								
Lab Title		Implementation of Half Adder, Full Adder, Sample Boolean Expression in Verilog HDL & FPGA Board.								
Submitted by										
Names		Registration Number								
Sumaira Safeer		FA22-BCE-019								
Rubrics name & number							Marks			
							In-Lab	Post-Lab		
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.									
	R3: Interpretation of Subject Knowledge: Ability to interpret and explain mathematical and/or visual forms, including equations, diagrams, graphics, figures, and tables.									
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.									
Design	R7: Implementing Design Strategy: Ability to execute a solution taking into consideration design requirements and pertinent contextual elements. [Block Diagram/Flow chart/Circuit Diagram]									
	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.									
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.									
	R11: Tools Evaluation: Ability to simulate the experiment and them using hardware tools to verify The results.									
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.									
	R13: Management of Teamwork: Ability to appreciate, understand and work with multidisciplinary team members.									
Rubrics to follow										
Rubrics	R2	R3	R5	R6	R7	R8	R9	R11	R12	R13
In-Lab										
Post-Lab										

LAB: 03

1. Objectives:

The objective of this lab is to get familiar with

- Implementation of half, and full adder using Verilog HDL and physically implementing that on Spartan6.
- Implementation of Verilog module that implements a Boolean expression using logic gates.

2. Lab Tasks:

Task#1:

Half Adder

- **Verilog Code**

```
module halfAdder(sum, cout, a, b);  
input a,b;  
output sum, cout;  
xor(sum, a, b);  
and(cout, a, b);  
endmodule
```

- **Verilog Test Bench**

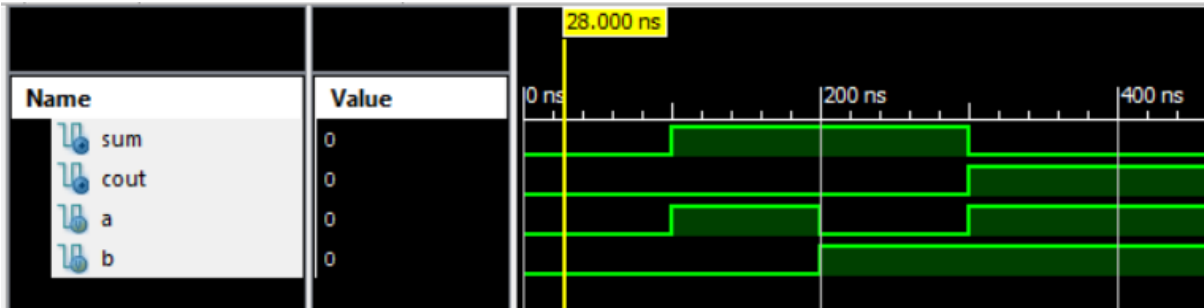
```
module halfAdderTest;  
    reg a;  
    reg b;  
    wire sum;  
    wire cout;  
    halfAdder uut (  
        .sum(sum),  
        .cout(cout),  
        .a(a),  
        .b(b)  
    );  
    initial begin  
        a = 0;  
        b = 0;  
        #100;  
        a = 1;  
        b = 0;  
        #100;  
        a = 0;
```

```

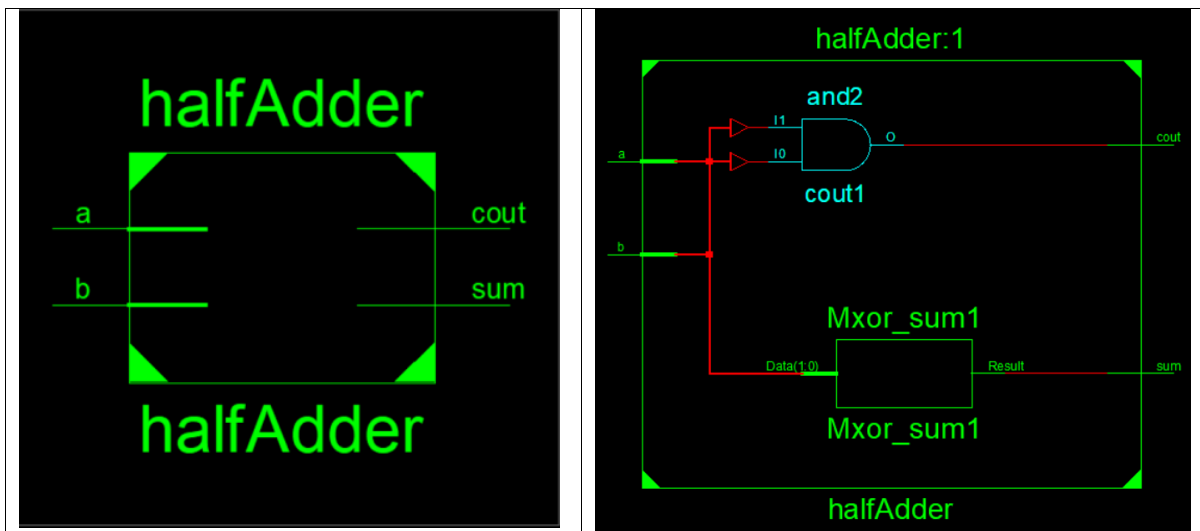
        b = 1;
        #100;
        a = 1;
        b = 1;
    end
endmodule

```

- Waveform



RTL Diagram



Task #2:

Full Adder:

- Verilog Code

```

module fullAdder(carry, sum, a, b, cin);
input a, b, cin;
output carry, sum;
wire w1, w2, w3;
xor X1(w1, a, b), X2(sum, w1, cin);
and A1(w2, a, b), A2(w3, w1, cin);
or(carry, w3, w2);

```

```
endmodule
```

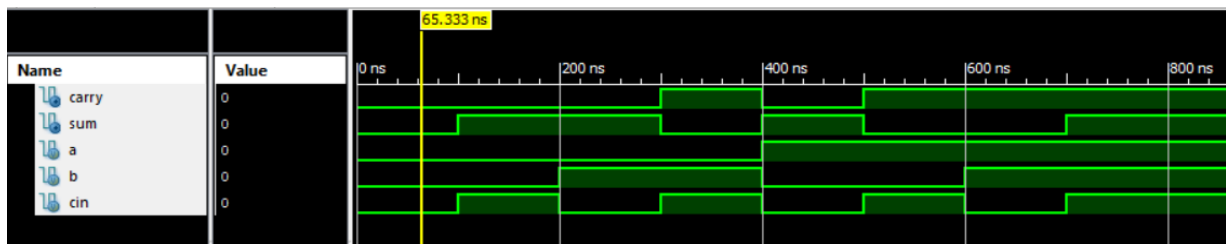
- **Verilog Test Bench**

```
module fullAdderTest;
    reg a;
    reg b;
    reg cin;
    wire carry;
    wire sum;
    fullAdder uut (
        .carry(carry),
        .sum(sum),
        .a(a),
        .b(b),
        .cin(cin)
    );

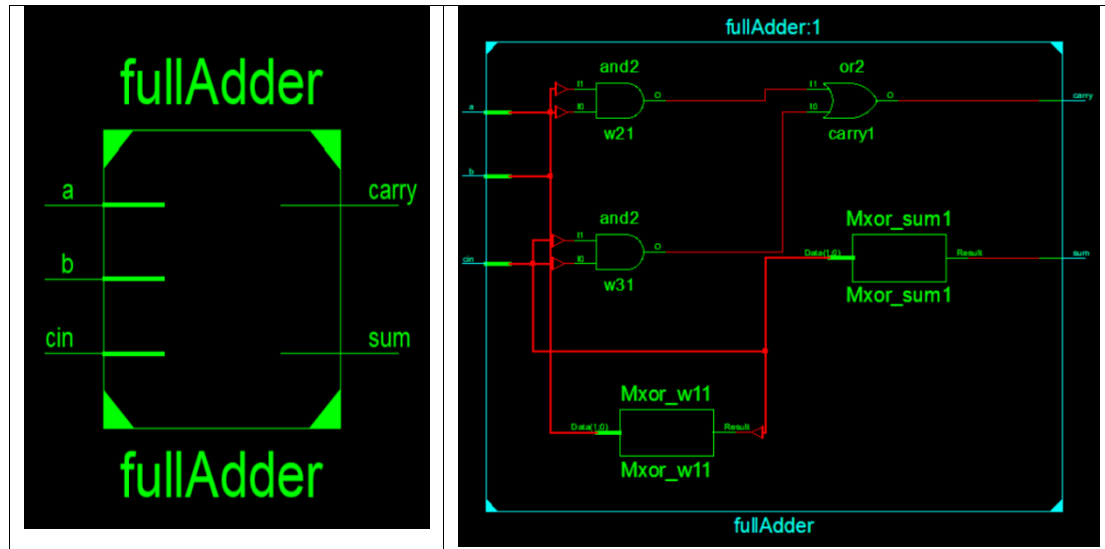
    a = 0; b = 0;
    cin = 0; #100;

    a = 0; b = 0;
    cin = 1; #100
    a = 0; b = 1;
    cin = 0; #100;
    a = 0; b = 1;
    cin = 1;
    #100;
    a = 1; b = 0;
    cin = 0;
    #100;
    a = 1; b = 0;
    cin = 1;
    #100;
    a = 1; b = 1;
    cin = 0;
    #100;
    a = 1; b = 1;
    cin = 1;
    end
endmodule
```

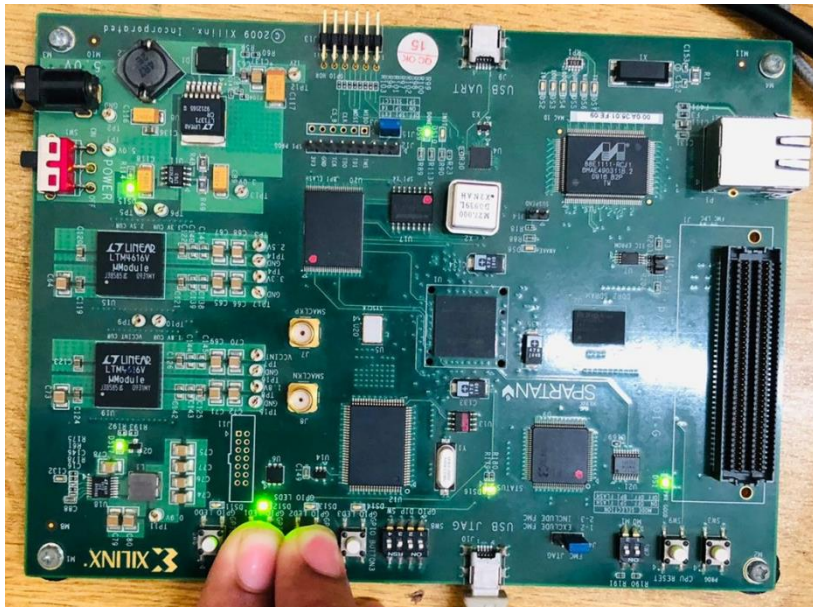
- Waveform



- RTL Diagram



- Hardware Implementation



Task#2:

- Verilog Code

```
module booleanExpression(y, x1, x2, x3, x4, x5);
```

```
input x1, x2, x3, x4, x5;
output y;
wire w1, w2;
and(w1, x1, x2);
nand(w2, x3, x4, x5);
nor(y, w1, w2);
endmodule
```

- **Verilog Test Bench**

```
module booleanExpressionTest;
    reg x1,x2, x3,x4,x5;
    wire y;

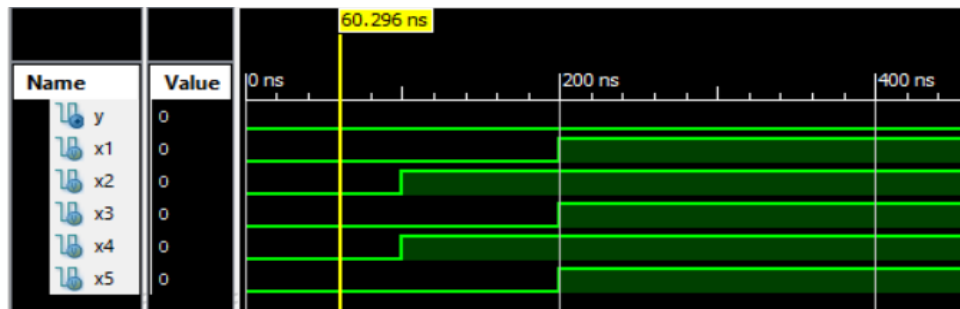
    booleanExpression uut (
        .y(y), .x1(x1),
        .x2(x2),
        .x3(x3),
        .x4(x4),
        .x5(x5)
    );

    initial begin
        x1 = 0; x2 = 0;
        x3 = 0; x4 = 0;
        x5 = 0; #100;

        x1 = 0; x2 = 1;
        x3 = 0; x4 = 1;
        x5 = 0; #100;

        x1 = 1; x2 = 1;
        x3 = 1; x4 = 1;
        x5 = 1; end
endmodule
```

- **Waveform**



- RTL Diagram

